Automatic Race Timing

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Introduction

- A computer vision application designed for timing a race
- Recognises runners bibs when they cross the finish line
- Gives times in real time
- Does not require the runners to have any extra equipment

Features

- Easy Race Configuration
- Accurate Bib Recognition/Timing
- Live Update of Results

Implementation

- Frontend
- Backend
- Computer Vision Model

Frontend Walk-through

- User Login/Registration
- Race Creation
- Data Upload

Backend Walk-through

- Website database interactions
- Result upload from CV models
- Result upload to the Web Interface

Computer Vision Model Walk-through

- Camera Input
- Bib Detection/ OCR
- Timing Calculation

Operational Setup

- System Architecture
- Operational Considerations
- Localhost

Demo

Enjoy this demonstration of the product

Planned Work

- Multi Camera Setup
- University Hosting
- Video Saving